Introductory Mathematical Economics

Prelims - Set

Delhi School of Economics

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Set

Reading: Simon and Blume Appendix A1

Set

A **set** is a collection of elements. Let X be a set. If x is an element in X then $x \in X$, otherwise $x \notin X$.

A set with no element is called an **empty set** and denoted by $\emptyset$.

Set

$A \subseteq B$ (A is a **subset** of B): All elements of A are member of B. Note that $\emptyset \subseteq A$ for any set A.

$A \subset B$ (A is **strict subset** of B): $A \subseteq B$ and there is at least one x such that $x \in B$ but $x \notin A$

$A = B$ (A is **equal** to B): If $A \subseteq B$ and $B \subseteq A$ then $A = B$. That is, all elements in A and B are the same.

A^c (**Complement** of A): Suppose all sets under consideration are subset of a fixed set U (which can be context specific). $A^c = \{x \in U \mid x \notin A\}$.

Set Operations

Union of A and B: $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$

Intersection of A and B: $A \cap B = \{x \mid x \in A \text{ and } x \in B\}$

A **set minus** B: $A - B = \{x \mid x \in A \text{ and } x \notin B\}$

Family of Sets

Let Λ be a non-empty set
For each $\alpha \in \Lambda$, there is a set A_α.
Λ is called the index-set, which can be either finite or infinite.

The definition of union and intersection do not change

$$\bigcup_{\alpha\in\Lambda} A_\alpha = \left\{x \middle| x \in A_\alpha, \text{ for some } \alpha \in \Lambda\right\}$$

$$\bigcap_{\alpha\in\Lambda} A_\alpha = \left\{x \middle| x \in A_\alpha, \text{ for all } \alpha \in \Lambda\right\}$$

Family of Sets: Examples

Example 1:

$\Lambda = \{Mon, Tue, Wed, Thurs, Fri\}$

$A_\alpha = \{x | x \text{ is a website that you have visited on day } \alpha\}$

Note that we are interested in set operations on A_α not on Λ

Universal set is the set of all websites.

Example 2:

$\Lambda = \{n \mid n \text{ is a positive integer}\}$

$A_n = \left(-\frac{1}{n}, \frac{1}{n}\right)$, where $\left(-\frac{1}{n}, \frac{1}{n}\right)$ is an interval

Universal set is the set of all real numbers

Useful Results

Distributive laws:

$(i)\ B \cap \left(\bigcup_{\alpha \in \Lambda} A_\alpha \right) = \bigcup_{\alpha \in \Lambda} \left(B \cap A_\alpha \right)$

$(ii)\ B \cup \left(\bigcap_{\alpha \in \Lambda} A_\alpha \right) = \bigcap_{\alpha \in \Lambda} \left(B \cup A_\alpha \right)$

Useful Results

De Morgan's laws

(i) $\left(\bigcup_{\alpha\in\Lambda} A_\alpha\right)^c = \bigcap_{\alpha\in\Lambda} A_\alpha^c$

(ii) $\left(\bigcap_{\alpha\in\Lambda} A_\alpha\right)^c = \bigcup_{\alpha\in\Lambda} A_\alpha^c$

A proof

We shall prove $\left(\bigcap_{\alpha\in\Lambda} A_\alpha\right)^c = \bigcup_{\alpha\in\Lambda} A_\alpha^c$

We need to show - (a) $\left(\bigcap_{\alpha\in\Lambda} A_\alpha\right)^c \subseteq \bigcup_{\alpha\in\Lambda} A_\alpha^c$ and (b) $\bigcup_{\alpha\in\Lambda} A_\alpha^c \subseteq \left(\bigcap_{\alpha\in\Lambda} A_\alpha\right)^c$

Step (a)

Take any $y \in \left(\bigcap_{\alpha\in\Lambda} A_\alpha\right)^c$
$\Rightarrow y \notin \left(\bigcap_{\alpha\in\Lambda} A_\alpha\right)$ (by definition of complement)
$\Rightarrow y \notin A_\alpha$ for some $\alpha \in \Lambda$ (by definition of intersection)
$\Rightarrow y \in A_\alpha^c$ for α identified in the previous step (by definition of complement)
$\Rightarrow y \in \bigcup_{\alpha\in\Lambda} A_\alpha^c$ (by definition of union)

Step (b)

Take any $y \in \bigcup_{\alpha\in\Lambda} A_\alpha^c$
$\Rightarrow y \in A_\alpha^c$ for some $\alpha \in \Lambda$ (by definition of union)
$\Rightarrow y \notin A_\alpha$ for α identified in the previous step (by definition of complement)
$\Rightarrow y \notin \left(\bigcap_{\alpha\in\Lambda} A_\alpha\right)$ (by definition of intersection)
$\Rightarrow y \in \left(\bigcap_{\alpha\in\Lambda} A_\alpha\right)^c$ (by definition of complement)

Cardinality of a Set

Definition: Cardinality refers to the number of elements in a set. It measures the "size" of the set.

Finite Sets: Sets with a countable number of elements. The cardinality is a natural number.

- Example: $A = \{1, 2, 3\}$ has a cardinality of 3, denoted as $|A| = 3$.

Infinite Sets: Sets with an uncountable number of elements. The cardinality can be more complex.

- Example: The set of natural numbers $\mathbb{N} = \{1, 2, 3, \ldots\}$ has an infinite cardinality, often denoted as $|\mathbb{N}| = \infty$.

Cardinality of a Set: Types of Cardinality

Finite Cardinality: If a set A has n elements, where n is a non-negative integer, then the cardinality is n.

- Example: $B = \{a, b, c, d\}$ has $|B| = 4$.

Countably Infinite Cardinality: A set is countably infinite if its elements can be put into a one-to-one correspondence with the natural numbers.

- Example: The set of integers $\mathbb{Z}$ is countably infinite, denoted as $|\mathbb{Z}| = \infty$.

Uncountably Infinite Cardinality: A set is uncountably infinite if it is not possible to list its elements in a sequence that pairs each element with a natural number.

- Example: The set of real numbers $\mathbb{R}$ between 0 and 1 is uncountably infinite.

Metric Space

- A metric space is a pair (X, d) where $X \neq \varnothing$ and $d : X \times X \to \mathbb{R}$ satisfies:
  1. $d(x, y) \geq 0$ with equality iff $x = y$,
  2. $d(x, y) = d(y, x)$,
  3. $d(x, z) \leq d(x, y) + d(y, z)$.
- Open and closed balls formalize neighborhoods and are central to convergence and continuity.

Open Ball: Definition

Let (X, d) be a metric space, $x \in X$, $r > 0$. The open ball of radius r centered at x is

$$B(x, r) = \{y \in X : d(x, y) < r\}.$$

Intuition: all points strictly within distance r of the center.

Open Ball: Example 1 – Real line

On $\mathbb{R}$ with $d(x, y) = |x - y|$,

$$B(a, r) = \{y \in \mathbb{R} : |a - y| < r\} = (a - r, a + r).$$

So open balls are open intervals.

Open Ball: Example 2 – Euclidean plane

On $\mathbb{R}^2$ with Euclidean metric

$$d((x_1, y_1), (x_2, y_2)) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2},$$

$$B((0,0), r) = \{(x, y) : \sqrt{x^2 + y^2} < r\}.$$

This is the interior of the disk (boundary excluded).

Open Ball: Example 3 – Discrete metric

Discrete metric on set X:

$$d(x,y) = \begin{cases} 0, & x = y, \\ 1, & x \neq y. \end{cases}$$

If $r \leq 1$ then $B(x,r) = \{x\}$. If $r > 1$ then $B(x,r) = X$.

Closed Ball: Definition

For (X, d), $x \in X$, $r > 0$, the closed ball is

$$\overline{B}(x, r) = \{y \in X : d(x, y) \leq r\}.$$

This includes points at a distance exactly r.

Closed Ball: Example 1 – Real line

In $\mathbb{R}$ with $d(x, y) = |x - y|$,

$$\overline{B}(a, r) = [a - r, a + r].$$

Closed balls correspond to closed intervals.

Closed Ball: Example 2 – Euclidean plane

In $\mathbb{R}^2$,

$$\overline{B}((0,0),r) = \{(x,y) : \sqrt{x^2+y^2} \leq r\},$$

The closed disk including its boundary.

Closed Ball: Example 3 – Discrete metric

With the discrete metric,

$$\overline{B}(x,r) = \begin{cases} \{x\}, & 0 < r \leq 1, \\ X, & r > 1. \end{cases}$$

Convex Set: Definition

A set $C \subseteq \mathbb{R}^n$ is convex if for every $x, y \in C$ and every $\lambda \in [0, 1]$,

$$\lambda x + (1 - \lambda) y \in C.$$

Intuition: the line segment joining any two points in C lies in C.

Convex Set: Key properties and examples

- Intersections of convex sets are convex.
- Convex hull exists and is unique.
- Linear inequality sets $\{x : a^\top x \leq b\}$ are convex.
- Budget set $\{x \geq 0 : p^\top x \leq w\}$ is convex.